SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO1: *Apply object-oriented concepts and software development life cycle models to design structured and modular software systems.* *(BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Ramamoorthy

QUESTIONS: 12 **Total Marks:60**

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **NIRANJANA (192521370)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

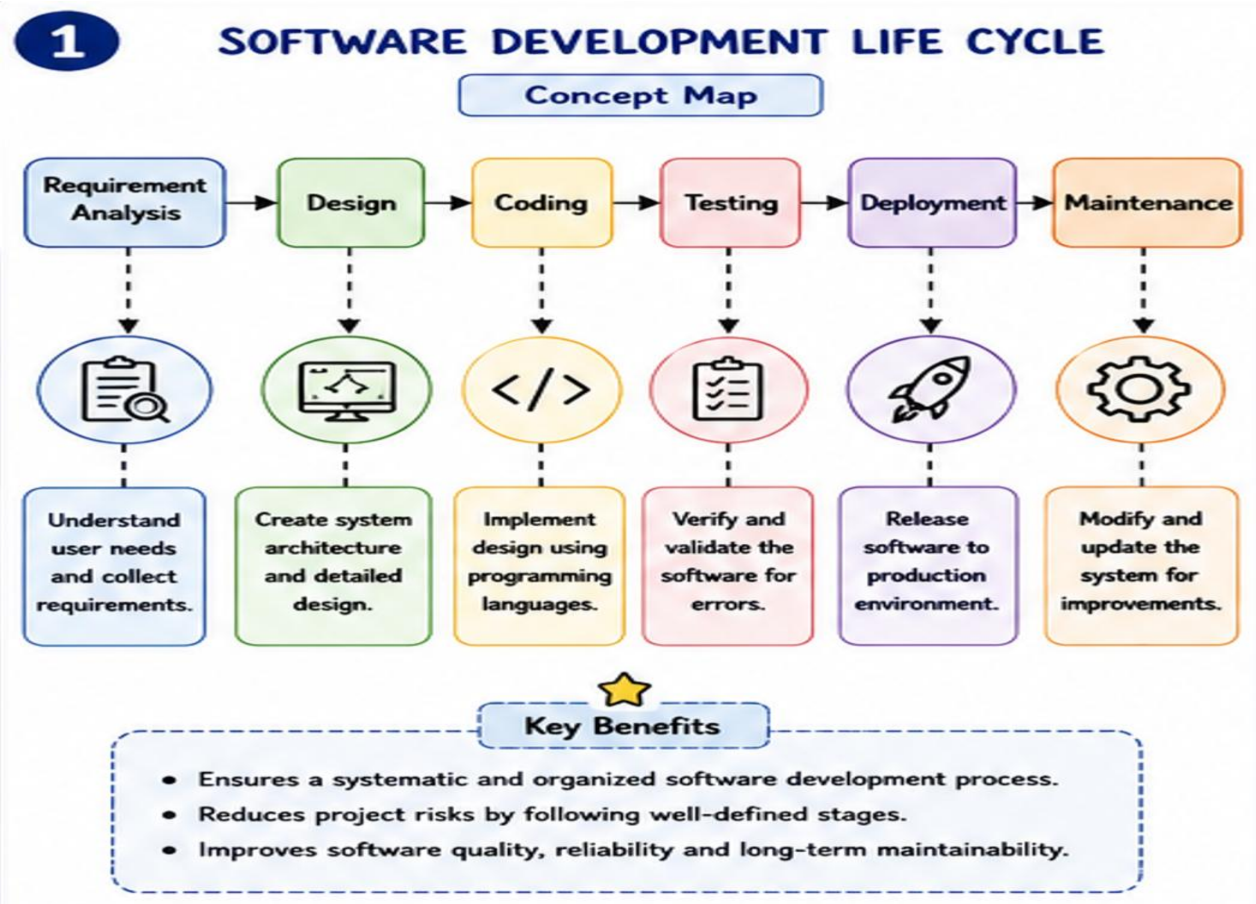
Annexure A

1. Construct a Concept Map

Given: Requirement Analysis → ______ → Coding → Testing → ______ → Maintenance

Answer: The missing concepts are:

- Design
- Deployment



Explanation:

- Design converts requirements into system architecture.
- Deployment releases the software to end users.

Analysis:

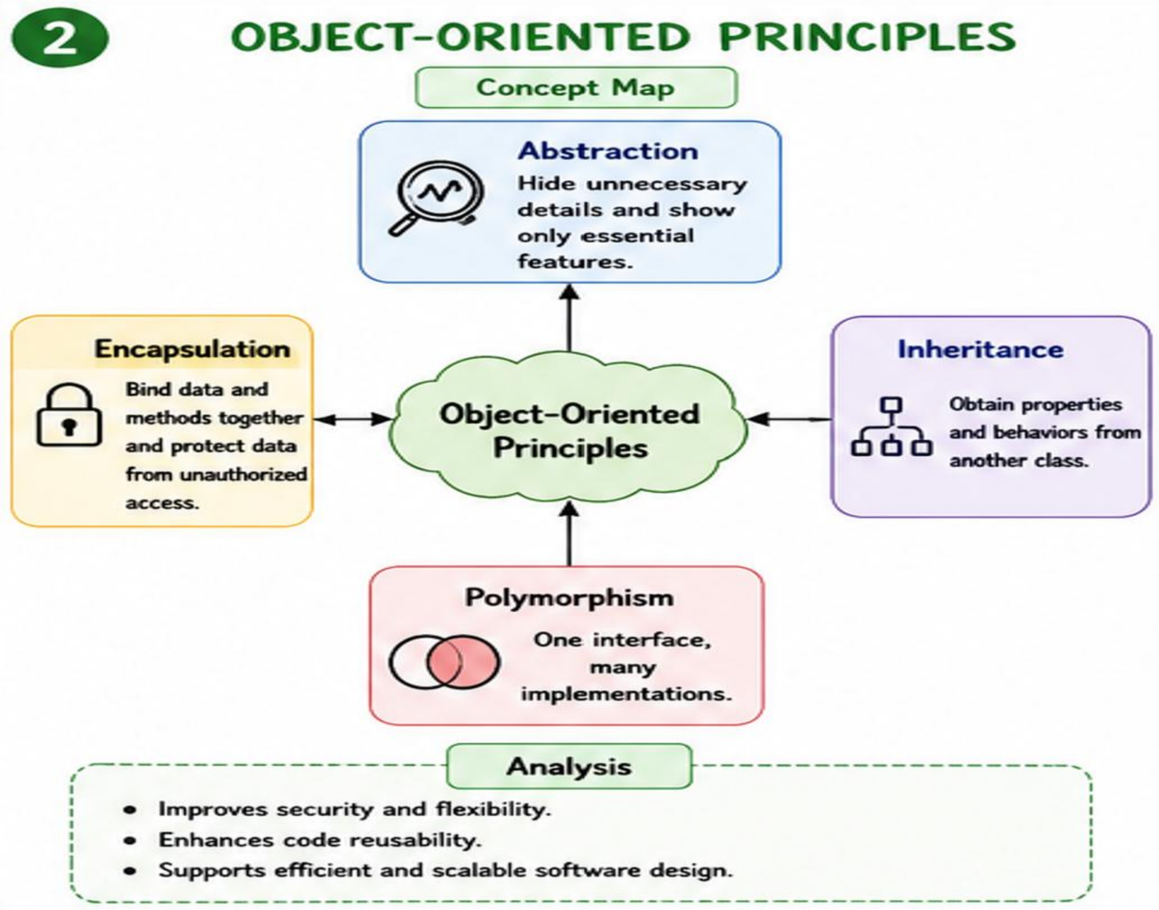
- Ensures systematic software development.
- Reduces project risks.
- Improves software quality and maintainability.

2. Construct a Concept Map

Given: Object-Oriented Principles → Abstraction → ______→ Inheritance → ______

Answer: The missing concepts are:

- Encapsulation
- Polymorphism



Explanation:

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviors.

Analysis:

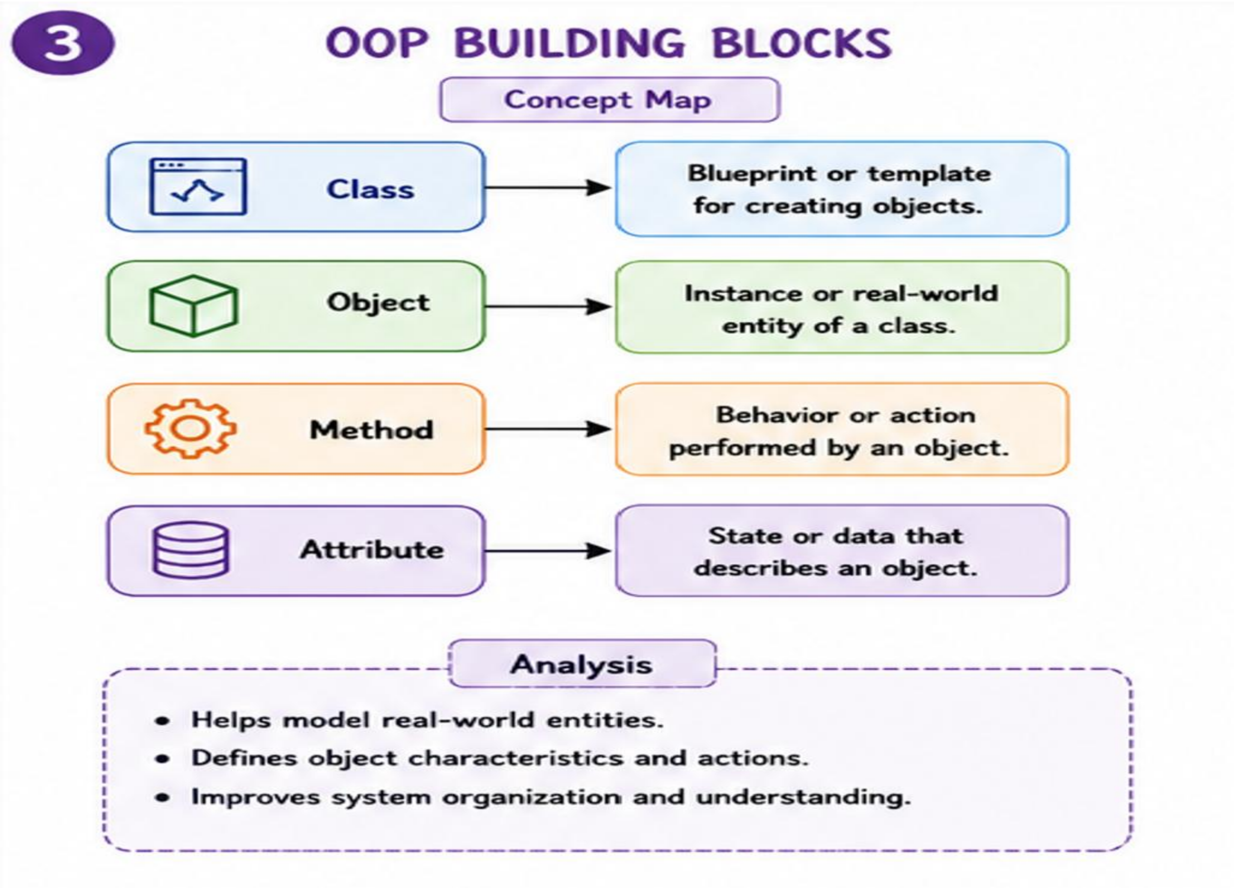
- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.

3. Construct a Concept Map

Given: Class → Blueprint Object → Instance Method → ______Attribute → ______

Answer: The missing concepts are:

- Behaviour
- State



Explanation:

- Methods define the behavior of objects.
- Attributes represent the state of objects.

Analysis:

- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.

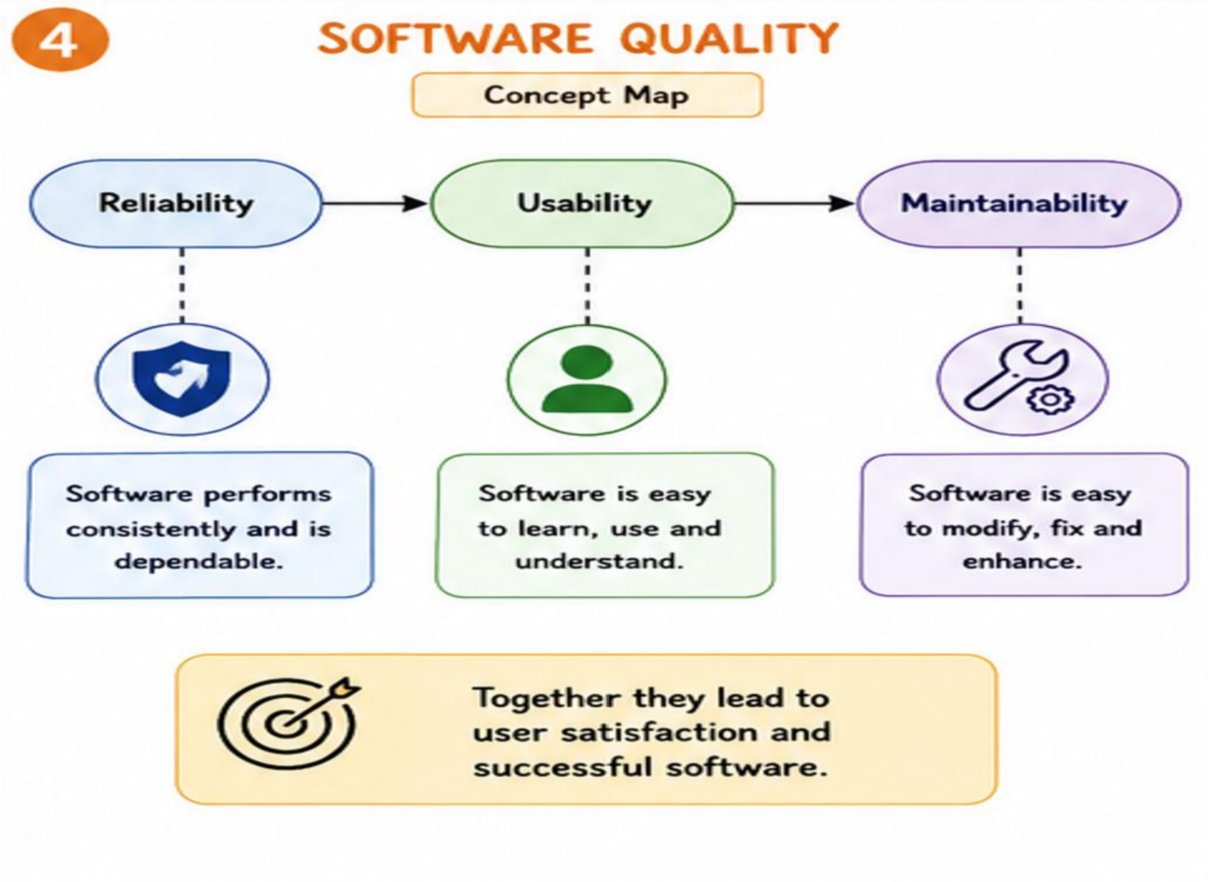
4. A Concept Map Shows

Given:

Software Quality→ Reliability→ ______→ Maintainability

Answer: The missing concept is:

- Usability



Explanation:

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

Analysis:

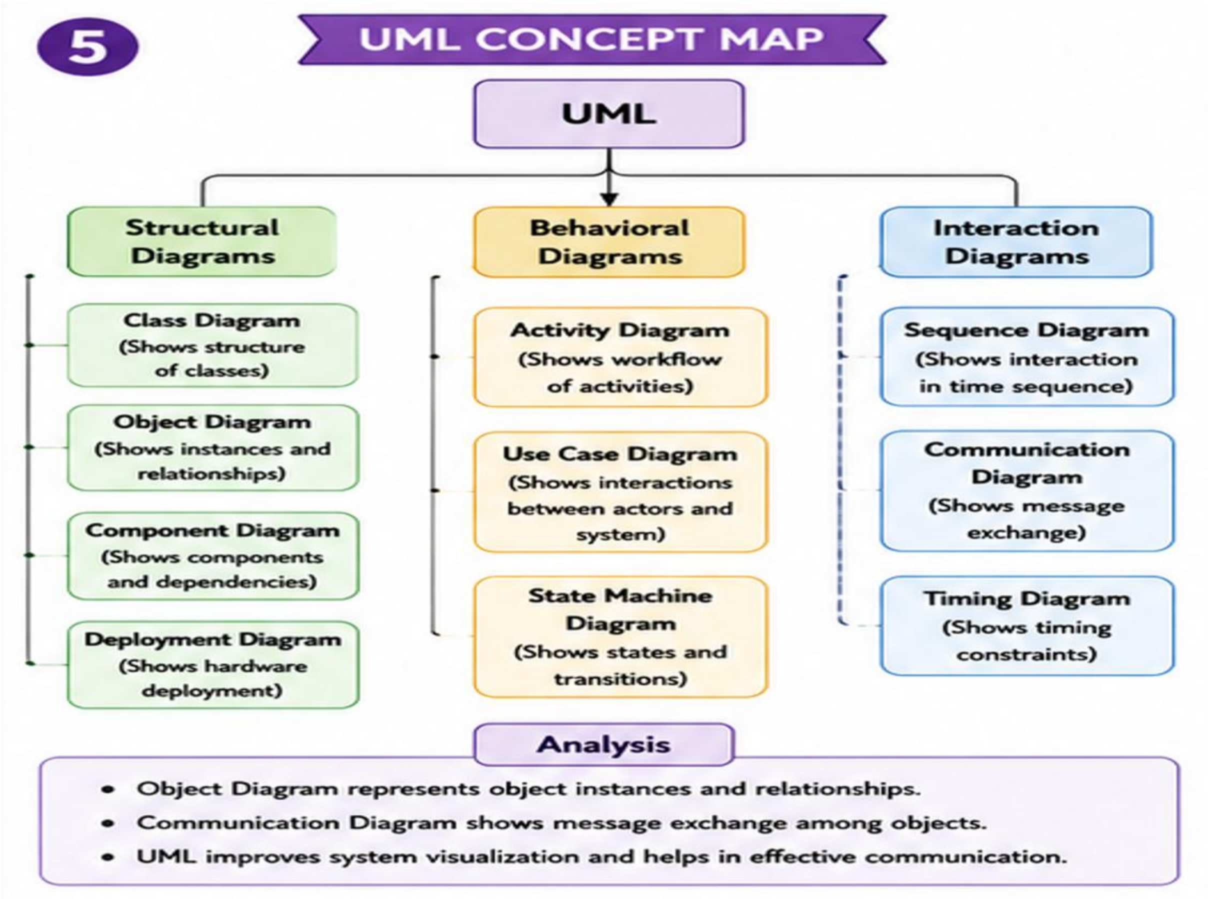
- Improves user satisfaction.
- Enhances product acceptance.
- Contributes to software success.

5. UML Concept Map

Given: UML → Structural Diagrams → ______ → Behavioral Diagrams → Activity Diagram → Interaction Diagrams → ______

Answer: The missing concepts are:

- Object Diagram
- Communication Diagram



Explanation:

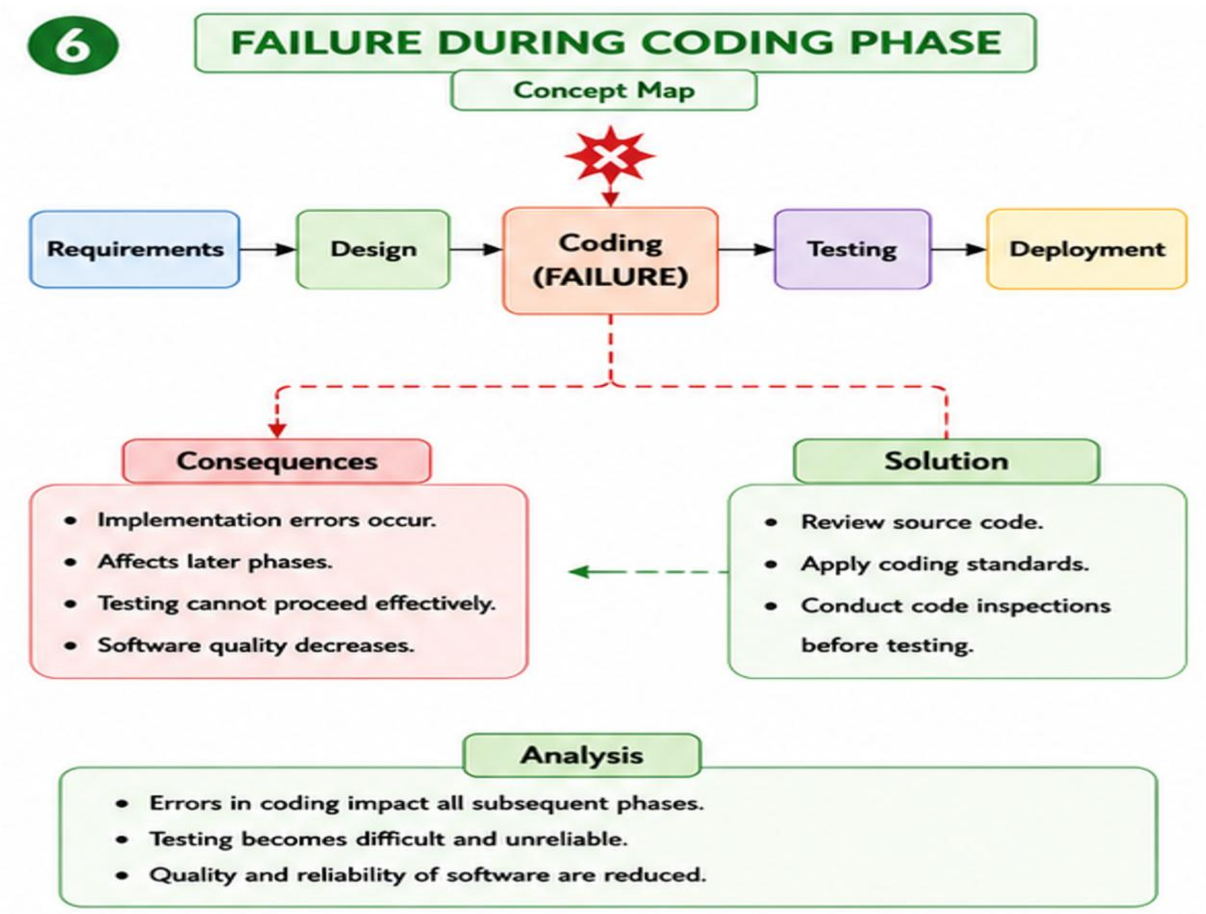
- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

Analysis:

- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.

6. Given:

Requirements ✔ → Design ✔ → Coding ✖ → Testing ✖ → Deployment ✖



Explanation:

- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

Analysis:

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

Solution:

- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.

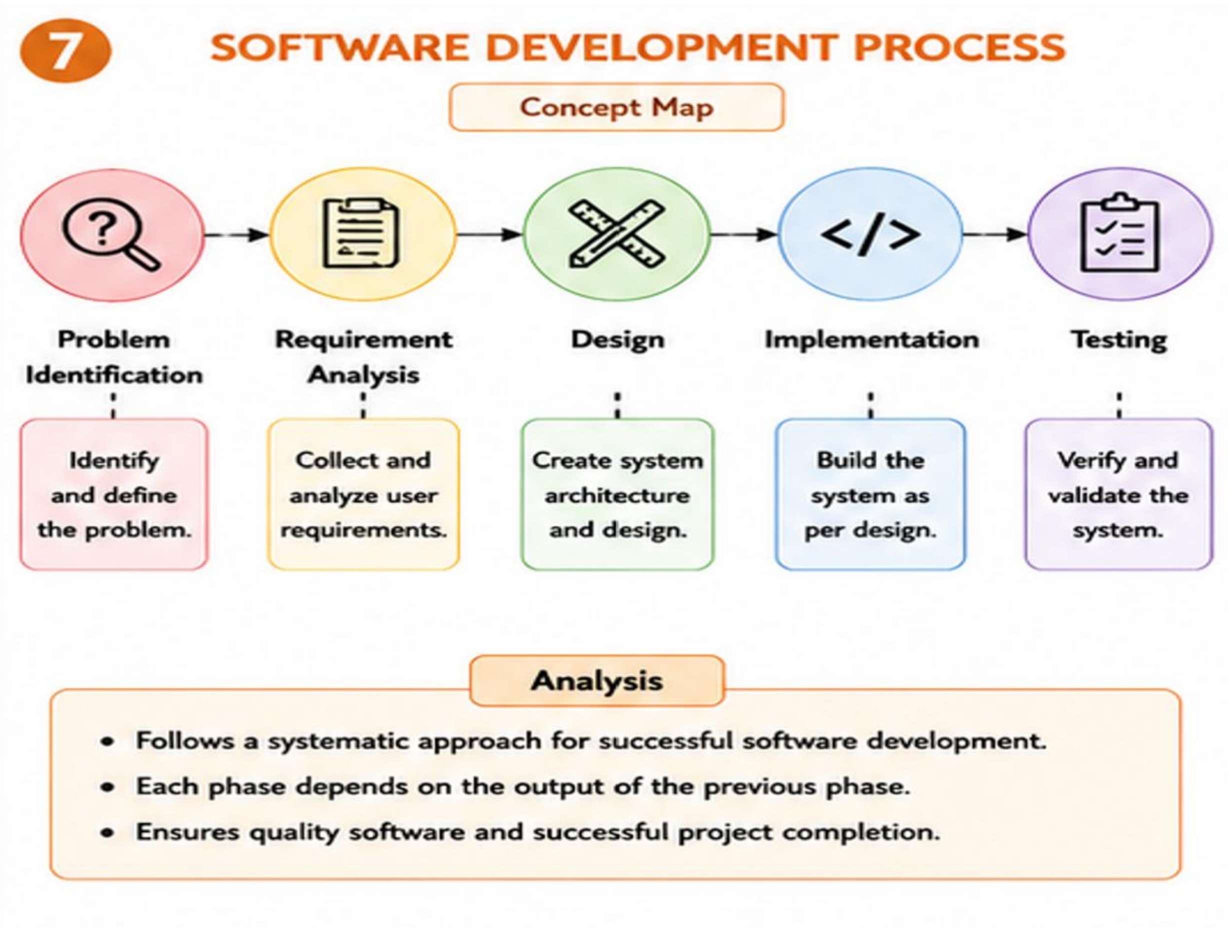
7. Construct a Concept Map

Given:

Problem Identification → ______ → Design → Implementation → ______

Answer: The missing concepts are:

- Requirement Analysis
- Testing



Explanation:

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

Analysis:

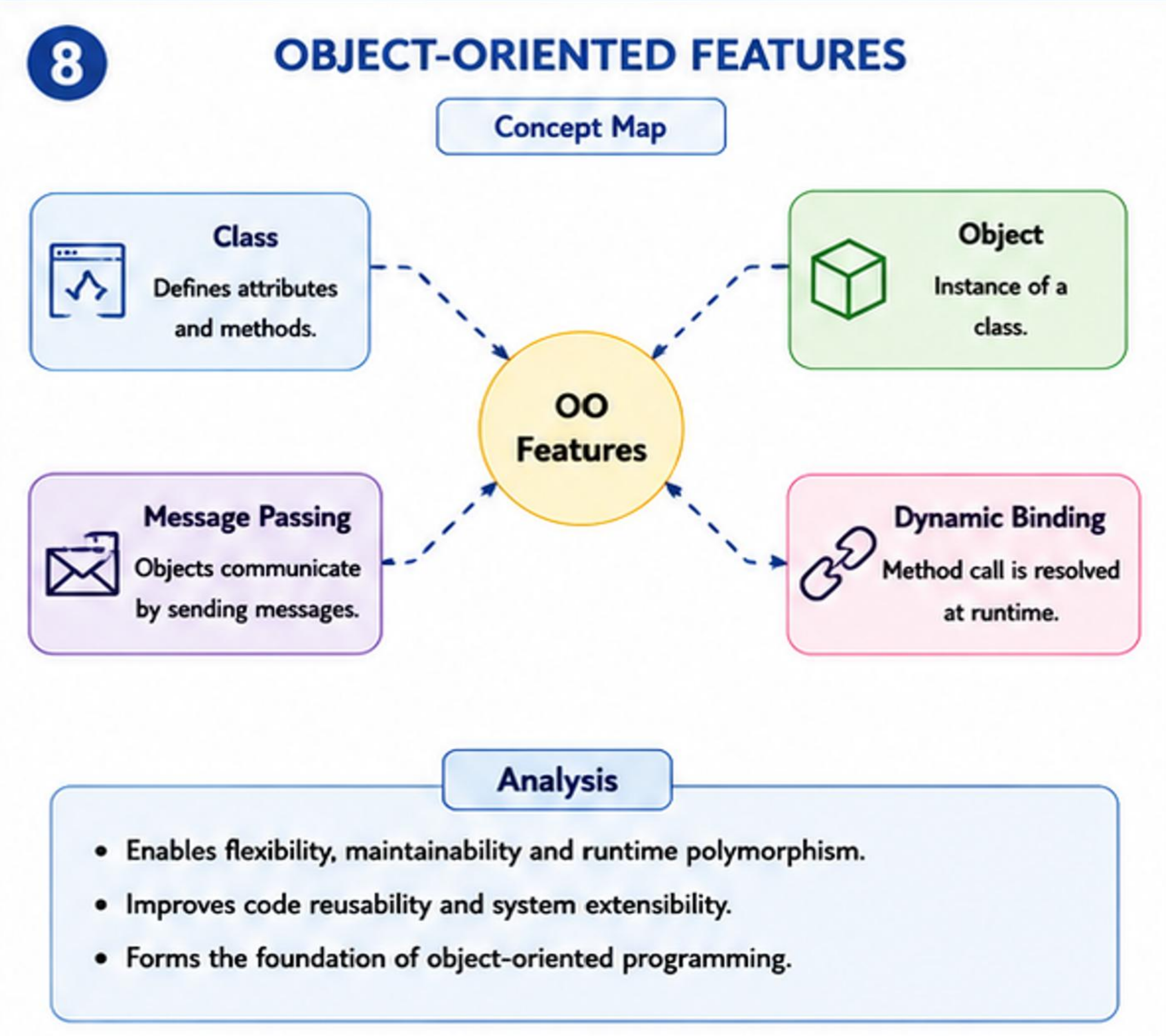
- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.

8. Construct a Concept Map

Given: Object-Oriented Features → Class → ______→ Message Passing→ ______

Answer: The missing concepts are:

- Object
- Dynamic Binding



Explanation:

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

Analysis:

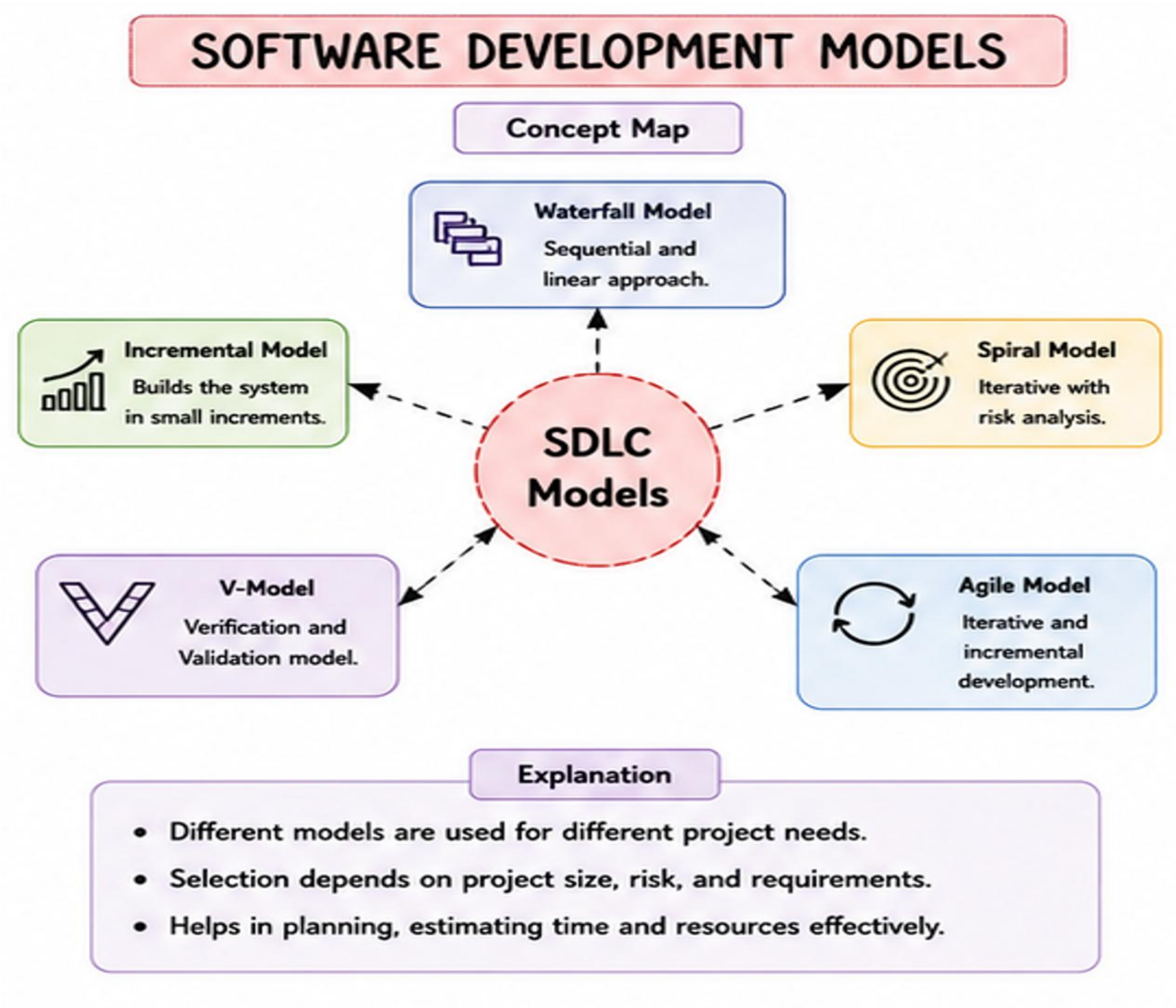
- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism

9. Construct a Concept Map

Given: Software Development Models→ Waterfall→ ______→ Spiral→ ______

Answer: The missing concepts are:

- Incremental Model
- Agile Model



Explanation:

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

Analysis:

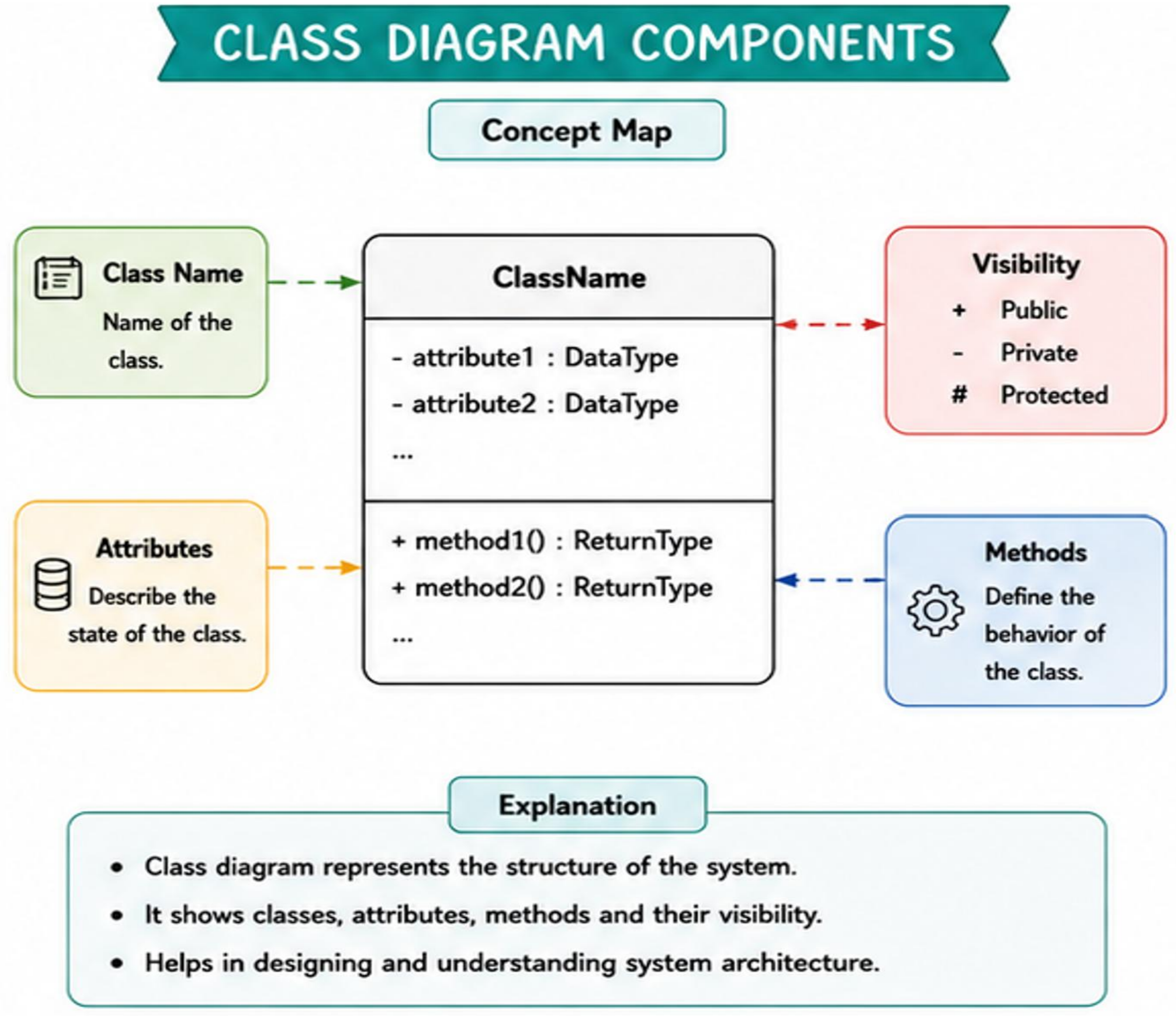
- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction

10. Construct a Concept Map

Given: Class Diagram → Class → ______→ Association→ ______

Answer: The missing concepts are:

- Attribute
- Method



Explanation:

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

Analysis:

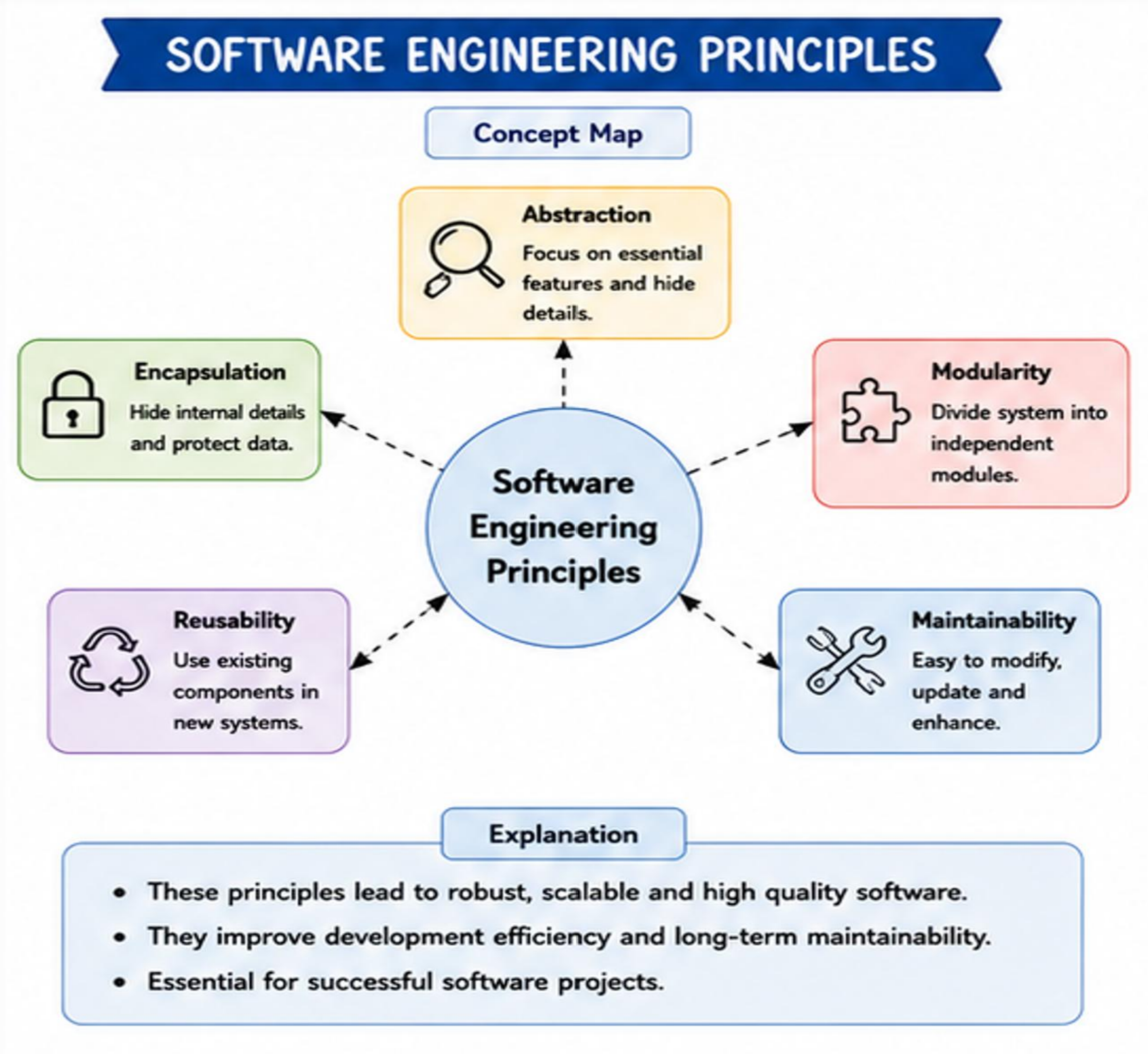
- Represents object structure effectively.
- Improves software documentation.
- Supports object-oriented

11. Construct a Concept Map

Given: Software Engineering Principles → Modularity→ ______→ Reusability→ ______

Answer: The missing concepts are:

- Maintainability
- Scalability



Explanation:

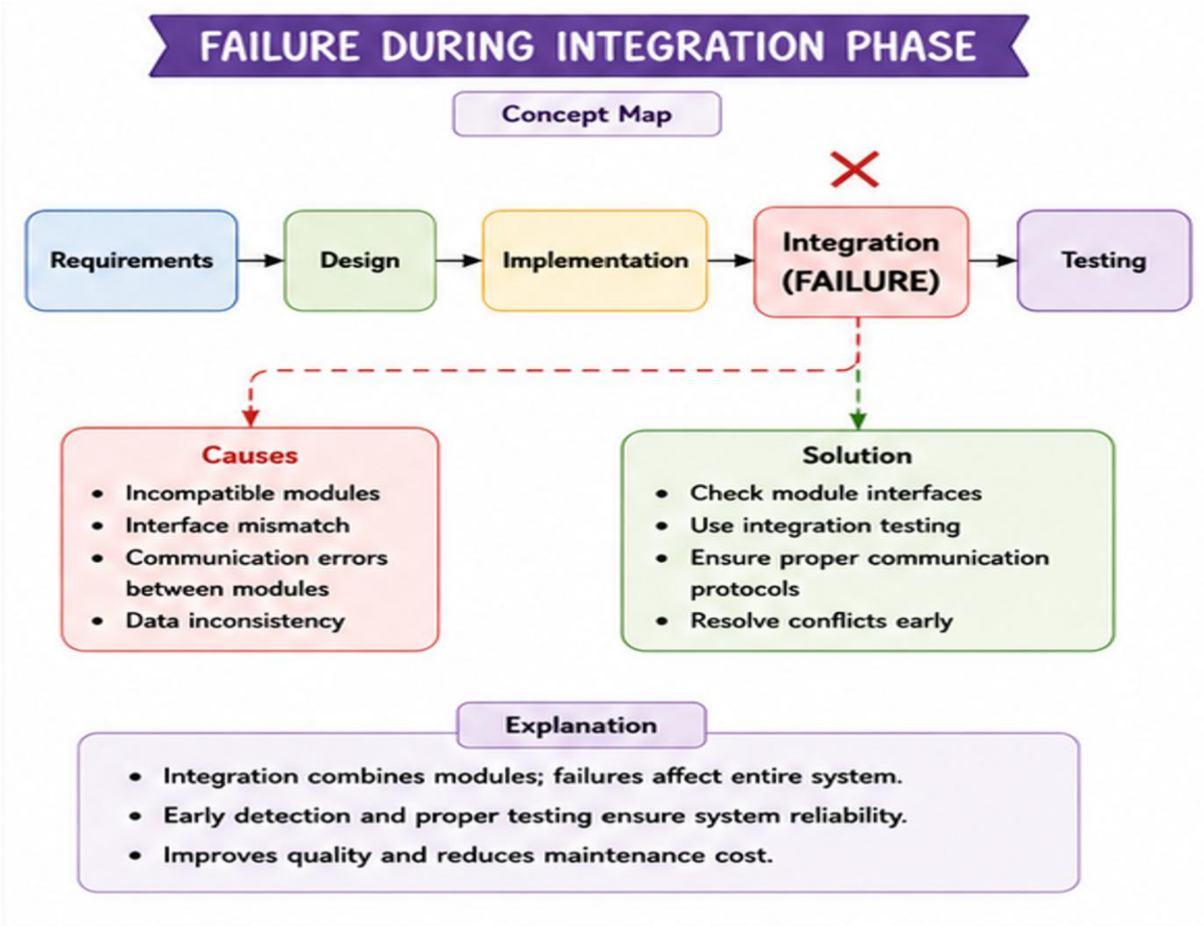
- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

Analysis:

- Improves software performance.
- Reduces maintenance effort.
- Supports long-term software evolution.

12. Given:

Requirement Analysis ✔ → System Design ✔ → Coding ✔ → Integration ✘ → Deployment ✘



Explanation:

- Failure occurs during the Integration phase.
- Individual modules fail to communicate correctly.

Analysis:

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

Solution:

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.

CONCEPT MAPPING

RUBRICS FOR CALCULATION

CONCEPT MAPPING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand